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Cite this: *Phys. Chem. Chem. Phys.*, 2022, **24**, 20073

Magnetic deep eutectic solvents: formation and properties†

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Deep eutectic solvents (DESs) are well-known as novel solvents due to their unique properties, which are indispensable for the development of green chemistry in the future. $CoCl_2{\cdot}6H_2O$ and $NiCl_2{\cdot}6H_2O$-based DESs, which could be called magnetic DESs (MDESs) for short in view of their responsive behavior to an external magnetic field, have been widely used in many industrial applications, such as biomass treatment, electrolytes, and material preparation. For better application and full-scale development of these MDESs in various fields, eleven MDESs were prepared in this work by using $CoCl_2{\cdot}6H_2O$ and $NiCl_2{\cdot}6H_2O$ as hydrogen bond acceptors (HBAs) with alcohols, carboxylic acids and amides as hydrogen bond donors (HBDs), respectively. The intermolecular interactions between the components of MDESs were characterized *via* FTIR, ^{1}H NMR and DSC analysis. In addition, physicochemical properties including density, viscosity, conductivity, ionicity, pH values, surface tension, thermostability and solvatochromic parameters were investigated. All MDESs exhibit acid characteristics and have good conductivity comparable with ionic liquids (ILs) and other DESs used for electrolytes. The results show that stronger H-bonding networks in Ni-based MDESs make them have higher density, viscosity, polarity and surface tension values than Co-based MDESs. Moreover, all prepared MDESs possess a good conductivity behavior which could be comparable to that of common organic solvents and ILs. According to this work, we could better comprehend the behavior of Co/Ni-based MDESs and choose the appropriate one for particular applications.

Received 6th April 2022,
Accepted 31st July 2022

DOI: 10.1039/d2cp01592g

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Introduction

Green chemistry is a significant research branch of future chemical industry development, which caters to the idea of being environmentally friendly and sustainable.[1] Over 70% of chemical engineering processes involve solvents, most of which are organic solvents at present. As is known to us all, organic solvents are toxic, flammable and easily volatile in general, causing a negative influence on the human body and natural environment. Consequently, eco-friendly, less toxic and safe novel alternative solvents are extremely in demand. In recent years, deep eutectic solvents (DESs), as emerging green solvents, have attracted more attention given their excellent properties of low cost, easy preparation, negligible vapor pressure and so on,[2–4] and they were first introduced by Abbot *et al.* in 2003.[5] Generally, they consist of a hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) and present a lower melting point or glass transition temperature than pure components.[6,7] More specifically, Abbott *et al.* have divided DESs into four types:[5] "Type I" DESs are formed with metal halides and quaternary ammonium salts (*e.g.* ChCl: $ZnCl_2$), "Type II" DESs are hydrated metal halides and quaternary ammonium salts (*e.g.* ChCl: $CrCl{\cdot}6H_2O$), "Type III" DESs are made of ionic-molecular eutectic mixtures (*e.g.* ChCl: urea), and "Type IV" DESs are mixtures of metal salts and molecular components (*e.g.* $ZnCl_2$: acetamide). On account of the designability and tunability of DESs, they have been applied in various applications, such as in extraction and separation,[8,9] biomass treatment,[10–12] material preparation,[13,14] and so on.

In recent years, "Type II" DESs have received a lot of attention in the materials field. Hydrated metal halides are regarded as economically and technically promising components to form DESs for industrial processes.[15] Specifically, Wang *et al.* have verified that the DES of $FeCl_3{\cdot}6H_2O$:urea could act as a catholyte, which exhibits a good reversibility and higher battery capacity.[16] A DES composed of $FeCl_3{\cdot}6H_2O$ and ethylene glycol has been reported to catalyze the conversion of cellulose, and it acts not only as a catalyst but a separation medium.[17]

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† Electronic supplementary information (ESI) available. See DOI: https://doi.org/10.1039/d1cp01592g

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Zn-based DESs have been prepared and used for PET degradation reactions.[18] Moreover, the Ni/Co-based DESs were favored to be used as precursors to prepare catalyst and adsorbent materials in view of the special solvent environment.[19–21] Furthermore, they manifested a notable performance in gas separation based on the coordination ability of metal ions as well.[22,23] This type of DES typically includes a magnetic transition metal, such as Fe(III), Co(II), Ni(II), Mn(II) and Zn(II), called magnetic DESs (MDESs) here. The physicochemical properties of MDESs are important to help us to understand their features and have a better application in various fields.[24,25] However, there are few studies on Co and Ni-based MDESs. Herein, eleven Co and Ni-based MDESs by using $CoCl_2{\cdot}6H_2O$ and $NiCl_2{\cdot}6H_2O$ act as HBA components, and alcohols, carboxylic acids and amides as HBD components were designed, prepared and characterized. Then, a comprehensive and systematic investigation on the properties and the intermolecular interactions between the components of the formed MDESs characterized by FTIR, ^{1}H NMR and DSC analysis were carried out. Temperature-dependent physicochemical properties such as density, viscosity and conductivity were performed at 25, 35, 45, 55, 65, and 75 °C. pH values, surface tension and solvatochromic parameters were determined at room temperature. In addition, a Walden plot was applied to determine the ionicity. The results obtained would reveal the characteristics of these prepared MDESs as potential industrial solvents.

Experimental

Materials

Cobalt(II) chloride hexahydrate ($CoCl_2{\cdot}6H_2O$, 98%), nickel(II) chloride hexahydrate ($NiCl_2{\cdot}6H_2O$, 98%), xylitol (Xyl, 98%), and acetamide (Ace, 99%) were all obtained from Aladdin Biochemical Technology Co., Ltd (Shanghai, China). Ethylene glycol (EG, 99%) was purchased from InnoChem Science & Technology Co., Ltd (Beijing, China). L(+)-Lactic acid (LA, 90%) was acquired from Macklin Biochemical Co., Ltd (Shanghai, China). Glycerol (Gly, 99%) was bought from Beijing Tongguang Fine Chemical Co. LTD. Urea (99%) was supplied by Sinopharm Chemical Reagent Co., Ltd. Nile red and 4-nitroaniline were purchased from J&K Scientific Ltd. All chemicals were dried thoroughly before use.

Synthesis

The Co and Ni-based MDESs were prepared according to the reported methods.[26] $CoCl_2{\cdot}6H_2O$ and $NiCl_2{\cdot}6H_2O$ were used as HBAs, and some typical polyols, amides and acids were chosen as HBDs (Fig. 1). The MDESs were obtained by mixing the HBA and HBD with a certain molar ratio (mainly 1 : 2 in this work) under heating (70 °C) and stirring (500 rpm). After a moment, a clear dark-blue or dark-green homogeneous liquid could be observed except for $NiCl_2{\cdot}6H_2O$ with xylitol. The obtained MDESs were dried under the vacuum environment at 80 °C for 24 h. The water content of MDESs was determined to be below 100 ppm by Karl Fischer titration. All MDESs maintain a stable state at room temperature. The main characteristics of MDESs are listed in Table 1.

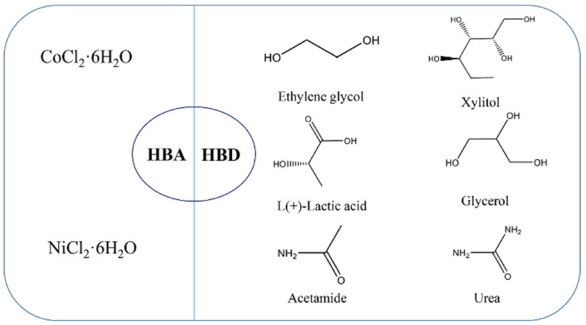


Fig. 1 The composition of all MDESs.

Characterization

The magnetic measurements were performed on a Model 7404 vibrating sample magnetometer (VSM, Lake Shore Corp, America). Fourier transform infrared (FT-IR) spectra were recorded on an FT-IR spectrometer (Tensor 27, Bruker) in the range of 4000 to 500 cm^{-1}. The nuclear magnetic resonance ^{1}H NMR spectra were recorded on a Bruker Avance 400 MHz spectrometer with D_6-dimethyl sulfoxide as the standard and the chemical shift data were later processed by the MestReNova Program. Differential scanning calorimetry (DSC) was performed using a DSC Q200 (TA Instruments, USA) with a heating rate of 5 °C min^{-1} under a nitrogen atmosphere. All the eutectic solvents were run in aluminum hermetic crucibles, which were cooled to −90 °C before heating up to 25 °C. TGA curves were measured by using a TGA Q4000 (PerkinElmer Instruments Inc.) in ramp mode (ramp 10 °C min^{-1} to 400 °C, platinum pan). The density measurements were carried out using an Anton Paar DMA 5000 M density meter with the accuracy of $\pm 5 \times 10^{-6}$ g cm^{-3}. And the viscosities for MDESs were measured with an Anton Paar AMVn viscosity meter using a falling ball technique, and the repeatability of the viscosity is $<0.1\%$. The ionic conductivity was measured using a conductivity instrument (DDS-307A, INESA). The pH values were measured by a pH-meter (PHS-3C, Shanghai INESA Scientific Instrument Co., Ltd, China). The surface tension of MDESs was measured using a tensiometer QBZY-1 (Wilhelmy platinum plate methods, Shanghai Fangrui Instrument Co., Ltd.) with the range of 0–600 mN m^{-1} and the accuracy of ±0.1 mN m^{-1}. The density, viscosity, conductivity, pH and surface tension values of each MDES were measured three times, and the average values are reported. For the measurement of Kamlet–Taft parameters, Nile red (NR) and 4-nitroaniline (NA) were dissolved in methanol with a concentration of 1.0×10^{-3} mol L^{-1} and then dried under vacuum at 40 °C. Then 20 μL of the dye solutions was added to a quartz cuvette containing 2.0 mL of the prepared MDES. A UV-visible spectrophotometer (Cary 5.0, Varian) was used to obtain the λ_{max} and duplicates of each MDES were measured at room temperature.

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Table 1 The characteristics of the MDESs

Name	Components (molar ratio)	Abbreviation	Appearance
MDES1	$CoCl_2{\cdot}6H_2O$/ethylene glycol (1 : 2)	$CoCl_2{\cdot}6H_2O$/EG	Clear dark-blue homogenous liquids
MDES2	$CoCl_2{\cdot}6H_2O$/xylitol (1 : 2)	$CoCl_2{\cdot}6H_2O$/Xyl	
MDES3	$CoCl_2{\cdot}6H_2O$/L(+)-lactic acid (1 : 6)	$CoCl_2{\cdot}6H_2O$/LA	
MDES4	$CoCl_2{\cdot}6H_2O$/glycerol (1 : 2)	$CoCl_2{\cdot}6H_2O$/Gly	
MDES5	$CoCl_2{\cdot}6H_2O$/acetamide (1 : 2)	$CoCl_2{\cdot}6H_2O$/Ace	
MDES6	$CoCl_2{\cdot}6H_2O$/urea (1 : 3)	$CoCl_2{\cdot}6H_2O$/urea	
MDES7	$NiCl_2{\cdot}6H_2O$/ethylene glycol (1 : 2)	$NiCl_2{\cdot}6H_2O$/EG	Clear dark-green homogenous liquids
MDES8	$NiCl_2{\cdot}6H_2O$/L(+)-lactic acid (1 : 6)	$NiCl_2{\cdot}6H_2O$/LA	
MDES9	$NiCl_2{\cdot}6H_2O$/glycerol (1 : 2)	$NiCl_2{\cdot}6H_2O$/Gly	
MDES10	$NiCl_2{\cdot}6H_2O$/acetamide (1 : 2)	$NiCl_2{\cdot}6H_2O$/Ace	
MDES11	$NiCl_2{\cdot}6H_2O$/urea (1 : 2)	$NiCl_2{\cdot}6H_2O$/urea	

Results and discussion

Magnetic properties of MDESs

The magnetic response behaviors of the prepared MDESs were investigated by vibrating sample magnetometry (VSM) at room temperature. As shown in Fig. 2a, all MDESs display a linear response, which indicates their paramagnetic behavior.[27] From a magnetic susceptibility point of view, the Co-based MDESs have a higher strength than Ni-based MDESs in general. But a higher content of HBD will weaken the magnetic response significantly. Moreover, the provided photographs in Fig. 2b show the static microdroplets of MDES6 and MDES11 both began to move in the presence of a magnetic field. Other MDESs showed a similar magnetic response. Briefly, the prepared MDESs present magnetic behavior.

Characterization of MDESs

The intermolecular interaction between components of the prepared MDESs was confirmed by applying Fourier transform infrared (FT-IR) spectroscopy. Fig. 3a and b present the relevant IR absorption bands of pure HBD components and MDESs. For the alcohols, the main characteristic peaks include the O–H stretching, O–H out-of-plane bending, and the C–O stretching vibration. As shown in Fig. 3a, the stretching vibration of O–H of the pure EG and Gly are ascribed to the peak at 3365 cm^{-1}. The out-of-plane bending vibration of O–H could be attributed to the peak at 1330 cm^{-1} and the corresponding C–O stretching vibration is at 1109 cm^{-1}. After formation of MDES1, MDES7, MDES4, and MDES9, the broadening of the O–H stretching peaks could be observed clearly. The corresponding absorption peaks appear at 3269, 3282, 3332, and 3282 cm^{-1}, respectively. Obviously, the results confirm the existence of a hydrogen-bonding network in these MDESs. Notably, the Ni-based MDESs have a broader O–H stretching vibration peak than the Co-based one, which might be caused by the stronger HB interaction between $NiCl_2{\cdot}6H_2O$ and alcohols. In the case of xylitol, the bands between 3450 and 3263 cm^{-1} appear as three small and sharp peaks which are assigned to the O–H stretching vibration. The C–O stretching vibration is at 1010 cm^{-1} with a sharp peak. However, it occurs just with a single wide peak at 3333 cm^{-1} in MDES2, and the C–O stretching vibration peak is at 1001 cm^{-1}. In particular, the peak shape of the C–O peak becomes so smooth and small that it almost disappears. The red shifts and changed peak shape could be attributed to the formation of hydrogen bonding between xylitol and $CoCl_2{\cdot}6H_2O$. As for the LA, the O–H stretching vibration is detected at 3370 cm^{-1}, and the C=O stretching vibration is observed at 1732 cm^{-1}. Besides, the O–H wagging vibration peak occurs at about 923 cm^{-1}, confirming the existence of carboxyl, as

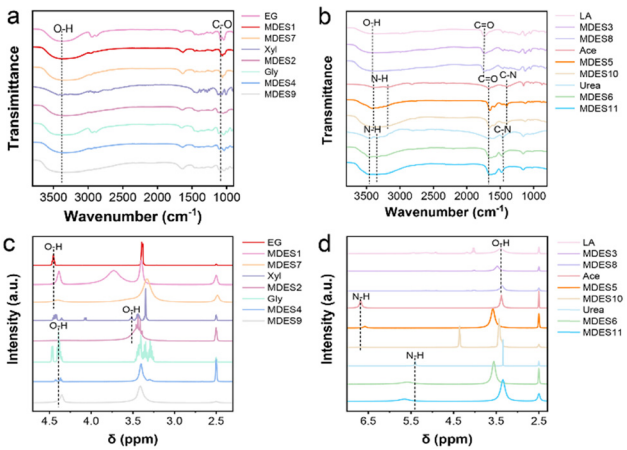


Fig. 3 (a) FT-IR spectra of pure alcohol HBD components and the relevant MDESs. (b) FT-IR spectra of pure acids, amides HBD components and the relevant MDESs. (c) ^{1}H NMR spectra of pure alcohols, acids and the relevant MDESs. (d) ^{1}H NMR spectra of pure amides and the relevant MDESs.

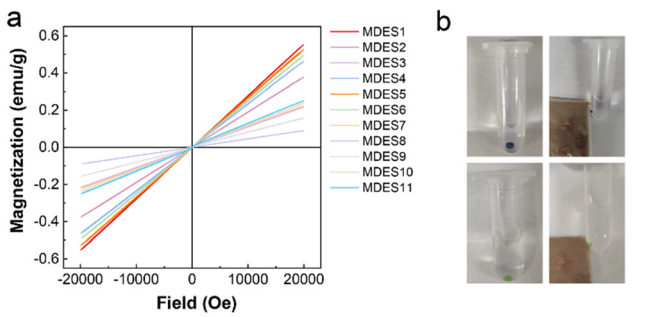


Fig. 2 (a) Field dependence of the magnetization of MDESs at room temperature. (b) Digital images of the status of MDESs without or with a magnet (top: MDES6; below: MDES11).

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depicted in Fig. 3b. In the spectra of the formed MDES3 and MDES8, the O–H stretching vibration peaks are both wider than those of pure LA and the C=O absorption peaks shift to a lower wavenumber. The phenomena prove the formation of hydrogen bonds between metal hydrated salts and LA. In terms of acetamide, the N–H asymmetrical stretching, N–H symmetrical stretching, in-plane N–H bending, the N–H wagging bending and C=O stretching vibration are ascribed to peaks at 3375 cm^{-1}, 3190 cm^{-1}, 1625 cm^{-1}, 1149 cm^{-1} and 1672 cm^{-1}, respectively. After the formation the MDESs, two sharp absorption peaks of N–H stretching vibrations in the range of 3540–3180 cm^{-1} become smooth and approximate a broad peak on the whole. Meanwhile, the C=O stretching vibrations generate bathochromic shifts and the broad N–H wagging bending vibration peak disappears. Thus, the weakening or disappearance of N–H stretching vibration peaks as well as the red shifts confirm the existence of a hydrogen-bonding network in MDESs. Compared with urea, the two N–H stretching vibration peaks are weak and the in-plane N–H bending vibration peaks disappear for MDES6 and MDES11. Similarly, the results prove the formation of hydrogen bonds between urea and hydrated metal halides. In a word, the formation of MDESs is to be credited to the hydrogen-bonding interaction among components.

Proton nuclear magnetic resonance (^{1}H NMR) is an effective technique to verify the HB interaction between components of DESs. In this work, the comparisons of chemical shifts of MDESs and related pure HBD components (solvent is D_6-DMSO, 2.50 ppm) are presented in Fig. 3c and d. Hydrogen bonding is a vital factor to impact the chemical shift of protons.[28] For alcohols and acids, O–H is a typical group and the active hydrogen proton could be used to form a H-bond. As shown in Fig. 3c, pure EG has one peak at 4.55 pm corresponding to the OH group. After adding the $CoCl_2{\cdot}6H_2O$ or $NiCl_2{\cdot}6H_2O$, the OH group in EG shifts to 4.38 ppm or 4.39 ppm. The chemical shift of the CH_2 group shifts up-field and the shape has a change in MDES1 and MDES7. This variation can also be seen in the spectra of Xyl, Gly and LA-based MDESs. In amide-based MDESs, N–H as a polar group is prone to form hydrogen bonding. Compared with pure acetamide, the chemical shift of the NH group shifts up-field in MDES5 and MDES10. In contrast, the N–H in urea-based MDESs shifts downfield in comparison with the pure urea component (Fig. 3d). There might be a stronger interaction between hydrated metal halides and urea. To sum up, the hydrogen-bonding network exists in all MDESs.

To gain further insight, a differential scanning calorimetry (DSC) experiment was carried out to confirm the formation of MDESs. As shown in Table 2, all the prepared MDESs present a low glass transition temperature (T_g) which is below −35 °C. Apparently, these values are lower than those of $CoCl_2{\cdot}6H_2O$ (86 °C), $NiCl_2{\cdot}6H_2O$ (80 °C), EG (−13 °C), Xyl (94 °C), LA (52 °C), Gly (20 °C), Ace (80 °C) and urea (135 °C), which is associated with the hydrogen bonding interactions between different components, indicating the formation of Co/Ni-based DESs. From this it seems clear that the prepared MDESs possess a good transport property for material preparation and chemical reaction. By extension, the thermal stability of MDESs was investigated by DTG-TGA analysis (shown in Fig. S1 and Table S1, ESI†). It could be found that all MDESs undergo a dehydration process firstly and then multi-stage decomposition. Among these, MDES2 has the highest thermal stability up to 281.6 °C followed by Gly-based MDESs, which is due to the low volatility and high thermal stability of their HBD components. Overall, the prepared MDESs have a relatively good thermo-stability compared with some other DESs.[29]

Table 2 Physicochemical properties (melting point, Kamlet–Taft parameters, pH and surface tension) of MDESs at 25 °C

MDES	T_g (°C)	α	β	π^*	pH	Surface tension ($mN{\cdot}m^{-1}$)
MDES1	−62.5	0.93	0.92	0.72	3.9	47.4
MDES2	−40.8	0.93	0.99	0.72	2.9	N/A
MDES3	−37.3	0.99	0.83	0.78	0.9	35.0
MDES4	−65.2	0.99	0.87	0.77	2.1	49.4
MDES5	−52.7	0.92	1.02	0.71	1.8	48.1
MDES6	−77.5	0.98	0.89	0.76	3.5	50.8
MDES7	−72.4	1.02	1.12	0.79	2.4	57.7
MDES8	−67.7	1.20	1.17	0.94	0.4	43.7
MDES9	−53.2	1.19	0.99	0.93	1.8	50.0
MDES10	−46.0	1.08	1.11	0.85	1.7	62.1
MDES11	−37.9	1.13	1.15	0.89	2.7	89.9

"N/A" denotes "not available". The mean values are given (except T_g (°C)) after three measurements.

Solvatochromic parameters

The solvation properties are critical to evaluate the application of solvents. Here, the Nile red λ_{max} and Kamlet–Taft (KT) parameters were measured to evaluate the solvent characteristics of MDESs. Firstly, Nile red λ_{max} is commonly used as a single parameter to measure the solvent polarity in view of its good solubility, low basicity and photochemical stability. As displayed in Fig. 4a, the Nile red shows a bathochromic shift in all MDESs, but the Ni-based MDESs have slightly superior shifts of λ_{max} than Co-based ones. Moreover, there are just subtle differences in λ_{max} between the MDESs composed with the same HBA. These results indicate that the HBA plays an instructive role in the whole polarity, which is consistent with a previous report.[30] Then, regarding the KT parameters, they include α, β, and π^*, with α showing the hydrogen-bond

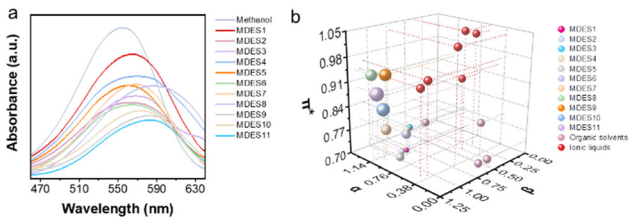


Fig. 4 (a) Absorption spectra of Nile red in methanol (used as a control solvent) and MDESs at room temperature. (b) The K–T solvatochromic parameters for MDESs. (Some typical organic solvents (acetonitrile, acetone, dichloromethane and methanol)[36] and ILs ([bmim][BF_4], [bmim][PF_6], [emim][NTf_2], [bmim]Oac, [bmim]O_2CEt and [bmim]$O_2CCH_2CO_2H$)[34] were reported as a comparison.)

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donating ability (acidity), β representing the hydrogen-bond accepting ability (basicity), and π^* referring to the polarity/polarizability of solvent, which are the most comprehensive and frequently used to measure the solvent properties quantitatively.[31–33] Here, these parameters were determined using Nile red (NR) and 4-nitroaniline (NA) dyes as probes and then calculated applying the following equations:[34]

$$\alpha = \frac{[19.9657 - 1.0241 \times \pi^* - \nu_{NR}]}{1.6078} \quad (1)$$

$$\beta = 11.134 - \frac{3580}{\lambda_{(NH_2)\,max}} - 1.125 \times \pi^* \quad (2)$$

$$\pi^* = \frac{[19.893 - \nu_{NR}]}{2.9912} \quad (3)$$

$$\nu_{NR} = \frac{1}{[\lambda_{(NR)\,max} \times 10^{-4}]} \quad (4)$$

On the basis of the equations, solvatochromic data for the MDESs at room temperature are listed in Table 2. From the results, the HBD acidity parameter α of MDESs ranges from 0.91 to 1.20. In general, when the MDESs have the same HBA, the acidity increases in the order amides < polyols < acid, which is the same with the variation of π^* values. The high acidic and polar nature might be caused by the carboxyl groups in the acid.[35] In contrast, the β values of acid-based MDESs are the lowest among all MDESs. On the whole, the Ni-based MDESs have slightly higher solvatochromic parameter values than those of Co-based ones, indicating the HBA has a major regulatory effect. Moreover, these parameters were measured in the Ni : Gly system with a different molar ratio (shown in Table S2, ESI†) as well. The comparison results indicate that the increase of –OH groups will decrease the acidity and polarity, but the β values are mainly affected by HBA, which are consistent with previous reports.[35] To better elucidate the solvation properties, a 3D plot was used to compare the prepared MDESs with typical organic solvents and ionic liquids (Fig. 4b). The results show that the α values of MDESs are close to those of methanol but superior to those of other selected solvents. Besides, all MDESs have higher β values as well. For the polarity, the MDESs could be comparable to the selected organic solvents and ionic liquids. Given the above, the MDESs have a fascinating solvation performance.

Density

Density is an essential and important physical property to understand the behavior of solvent. Meanwhile, it also influences the design and operation of processes in the environment and various industrial applications.[37] Thus, it is crucial to identify the behavior of density with respect to temperature. In this study, the density of the MDESs was measured at different temperatures (25, 35, 45, 55, 65, and 75 °C), and the results are depicted in Fig. 5a. It can be found that there is a negative linear correlation between density and temperature as

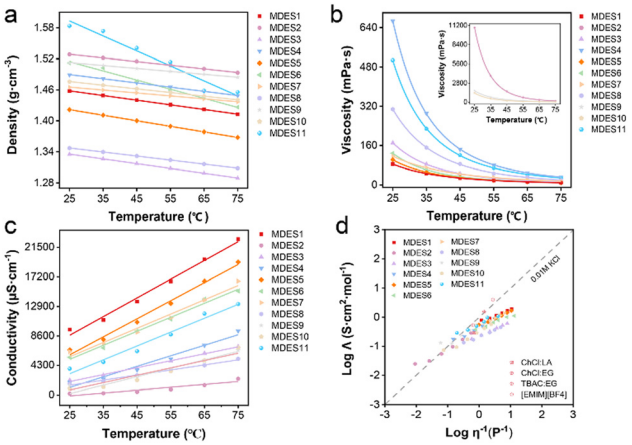


Fig. 5 (a) Density and (b) viscosity, fitted to the Vogel–Fulcher–Tammann (VFT) equation. (inset) The viscosities of MDES2, MDES9, and MDES10 from top to bottom and (c) conductivity of MDESs as a function of temperature. All values were tested at 25, 35, 45, 55, 65, and 75 °C. (d) Walden plot for the various DESs applied in this study (ChCl:LA, ChCl:EG, TBAC:EG, and [EMIM][BF_4] were reported as a comparison).

expected (the relevant fitting coefficients are given in Table S3, ESI†). This is due to the increased activity and mobility of these MDESs following the heat boost, which are equal to increasing the molar volume of the solvent. Thus, the density of the MDES decreases with the increase of temperature. All the recorded density values of the prepared MDESs are below 1.60 g cm^{-3}. The MDES11 has the highest density value (1.584 g cm^{-3}) at 25 °C among these MDESs. This might be caused by a stronger intermolecular interaction between $NiCl_2{\cdot}6H_2O$ and urea. In particular, the density values of MDES11 and MDES6 both have a sharp declining tendency when urea acts as a HBD. On the contrary, the variation in other MDESs is nearly consistent. This illustrates that the urea-based MDESs have a more intensive thermal expansion, which causes poor stability in the heating process. The minimum density value belongs to MDES3 (1.335 g cm^{-3}) at 25 °C, but this value is still higher than that of the majority of common organic solvents.[38] Moreover, for the MDESs with the same HBD components, the density value of Ni-based MDES is always greater than that for Co-based MDESs, indicating that HBA plays a crucial factor in it. For the Ni-based MDESs at 25 °C, there is an interesting rule for density values: Ni : urea (1.58 g cm^{-3}) > Ni : Gly (1.51 g cm^{-3}) > Ni : Ace (1.48 g cm^{-3}) > Ni : EG (1.47 g cm^{-3}) > Ni : LA (1.35 g cm^{-3}), which is suitable for Co-based MDESs (Fig. 5a). These results show that the density of MDESs is comparable to that of reported transition metal-based DESs, traditional ChCl-based DESs, and even *N*,*N*-diethyl-*N*-methyl-*N*-(2-methoxyethyl) ammonium iodide (DEME-TFSI, 1.42 g cm^{-3}).[39–41]

Viscosity

The meaning of viscosity is an internal friction of moving fluid. In other words, it describes the resistance of a liquid to flow. Thus, it is a vital factor influencing the fluidity and the rate of mass transport of a solvent. Here, the viscosity values of MDESs

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were studied within the same temperature range as the density measurement and the values are plotted against temperature in Fig. 5b. Besides, the viscosities were correlated by utilizing the modified Vogel–Fulcher–Tammann (VFT) equation and the relevant adjustable parameters are presented as well (see more in the ESI† and Table S4). As expected, the viscosity of each MDES decreases with increasing temperature exponentially. This indicates that the viscosity values of MDESs are highly sensitive toward temperature compared with density. In particular, the viscosity decreases rapidly during the initial warming process, and then it gradually approaches a balance value at higher temperature. This variation of viscosity is consistent with the reports.[42] It is noticeable that the viscosity of the studied MDES ranges from 84.3 to 10907.2 mPa s at 25 °C. Co:Xyl has the maximum viscosity value which is 6 to 100 orders of magnitude higher than that of the other MDESs. This could be attributed to the stronger hydrogen bonding network between $CoCl_2{\cdot}6H_2O$ and xylitol compared with other MDESs, which intensively influences the mobility of free species within MDES2. Generally, the sugar alcohol-based DESs possess a higher viscosity due to the existence of more robust 3D intermolecular hydrogen-bond networks.[43] Moreover, with $CoCl_2{\cdot}6H_2O$ as a HBA and alcohols as a HBD, the viscosity values increase in the following order: MDES1 < MDES4 < MDES2. This result is consistent with a previous report that more hydroxyl groups in DESs result a higher viscosity.[42] The extra hydroxyl groups promote a stronger hydrogen bonding interaction among components, which makes a viscous eutectic mixture. Moreover, this order indicates that the increase of chain lengths of the HBD further leads to the formation of a higher viscosity eutectic mixture. It could be explained in that higher molar mass side chains make DESs have stronger van der Waals forces, which results in insufficient mobility and high viscosity.[44] Specially, MDES4 has a far higher viscosity value compared to MDES1. This might be related to the mole ratio of the eutectic components $CoCl_2{\cdot}6H_2O$ with glycerol (1 : 3). In the same way, the viscosity values of Ni-based MDESs with alcohols as HBD are in the following order: MDES9 > MDES7. As a result, the structures of HBDs are an important factor in determining viscosity of MDES. Among the prepared MDESs, the viscosity values of Ni-based MDESs are higher than those of the Co-based ones with the same HBD components. This result shows that there is a stronger hydrogen bonding interaction between the components of Ni-based MDESs than in Co-based MDESs, which is consistent with the result of the FTIR experiment. This trend is compatible with the Mark–Houwink equation commonly used to describe the viscosity behavior of polymers.[45] The trend of the viscosity-temperature profile of these MDESs could be explained by the Hole Theory, which describes the influencing factors of the viscosity of ILs as well as DESs.[44] When heating the MDESs, the increased intramolecular energy speeds up the molecular motion and then the formed large hole will permit molecules to pass through each other easily, thereby decreasing the viscosity. In general, the viscosity values of MDESs are similar to those of ILs (50 to 10 000 mP s at 25 °C).

Table 3 Parameters for activation energies and the thermodynamic functions of activation of viscosity flow (kJ mol^{-1}). Coefficient of correlation (r) for fits to eqn (5) and (8) are also given

MDES	E_η	r	E_σ	$-r$	ΔH^*	$T\Delta S^*$	ΔG^*
MDES1	38.26	0.994	15.55	0.994	37.72	13.70	51.42
MDES2	70.95	0.998	43.39	0.975	70.55	33.62	104.2
MDES3	44.98	0.992	21.13	0.993	44.38	18.65	63.03
MDES4	52.47	0.993	29.22	0.990	51.99	22.61	74.60
MDES5	41.06	0.991	19.20	0.997	40.40	15.95	56.34
MDES6	47.32	0.992	18.31	0.987	46.29	21.60	67.89
MDES7	34.36	0.997	16.98	0.995	34.02	9.15	43.17
MDES8	46.58	0.994	20.81	0.994	46.08	18.85	64.93
MDES9	54.67	0.997	29.59	0.933	54.34	22.64	76.98
MDES10	59.95	0.993	39.62	0.984	59.53	28.84	88.37
MDES11	49.30	0.983	22.88	0.948	47.66	19.50	67.16

Furthermore, the temperature dependent viscosity plot for MDESs could be expressed by the following Arrhenius eqn (5), which describes the temperature dependence for non-associating electrolytes.

$$\ln \eta = \ln \eta_0 + \frac{E_\eta}{RT} \quad (5)$$

in which η is the viscosity, E_η is the activation energy for viscous flow, η_0 is a constant, and R is the gas constant. The E_η values are valuable for designing the low viscosity fluids, which are listed in Table 3. For better understanding the viscous flow, the thermodynamic functions of activation were calculated using Eyring's transition state theory. It can be expressed by the following eqn (6):

$$\ln\left(\frac{\eta V}{hN_A}\right) = \frac{\Delta G^*}{RT} \quad (6)$$

where h is Planck's constant, N_A is Avogadro's number, V is the molar volume of the MDES, and ΔG^* is the molar Gibbs energy. The value of ΔG^* could be calculated by applying formula (7):

$$\Delta G^* = \Delta H^* - T\Delta S^* \quad (7)$$

where ΔH^* and ΔS^* represent the enthalpy and entropy of activation of viscous flow, respectively. Combining with eqn (6) and (7), the absolute values of the thermodynamic functions of activation at 25 °C are obtained and presented in Table 3. The results reveal that the molar enthalpy is more important than the molar entropy for activation. It might state that the interaction is predominant over the structure for all studied MDESs in a viscous flow process.[25]

Conductivity and ionicity

At present, DESs play an important role as electrolyte materials for electrocatalysis, electroplating, and electrodeposition. Thus, the conductivity is necessary information for considering the application of MDESs. In general, the conductivity values increase following the increase in temperature because the kinetic energy from heat promotes the frequent collisions between molecules, thereby weakening the intermolecular forces of DES.[44] The conductivity values of MDESs were experimentally determined at different temperatures which are 25, 35, 45, 55, 65 and 75 °C, respectively. Fig. 5c shows

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the conductivity of MDESs as a function of temperature. It can be seen that a linear relationship appears between conductivity and temperature for all measured MDESs. At 25 °C, the conductivity values range from 209 to 9530 μS cm^{-1}. It is known that the conductivity of a DES is highly affected by its viscosity and ion concentration. It was reported that a viscous DES always gives a lower conductivity.[46] As hypothesized, MDES2 with the highest viscosity has the lowest conductivity value. This could be explained by the fact that the robust H-bonding network between $CoCl_2{\cdot}6H_2O$ and Xyl impedes the motion of ions. On the contrary, MDES1 has the highest conductivity compared with other MDESs because its higher ion mobility forms a larger void volume in the eutectic mixture. For alcohol donors, the conductive values are in the following order: MDES1 > MDES7 > MDES4 > MDES9 > MDES2. This result indicates that the number of hydroxyl groups of alcohols has an apparent effect on the conductivity of MDES. In detail, the diol-based MDES possesses fewer hydrogen bonding networks than the triol-based MDES, which leads to greater ion mobility and conductivity. Similarly, for the same system with different HBD content, the introduction of more –OH groups will decrease the conductivity (see Table S2, ESI†). In view of the close influence of viscosity to conductivity, the mostly Ni-based MDESs have lower conductivity values than Co-based MDESs. Compared with the reported $MnCl_2{\cdot}4H_2O$-based DESs,[46] the conductivity values of MDESs in this study are relatively higher than them. Analogous to the viscosity data, the conductivity of MDES varies with temperature according to eqn (8):

$$\ln \sigma = \ln \sigma_0 - \frac{E_\sigma}{RT} \quad (8)$$

where E_σ is the activation energy for conduction, T is the absolute temperature and σ_0 is a constant. The E_σ and correlation coefficients r are displayed in Table 3.

Now, ionicity is usually used to further understand the conductivity behavior of DES which reflects the effective portion of ions participating in conduction practically.[47] The Walden plot, a simple qualitative and semi-quantitative method, is applied to estimate the ionicity.[48] Here, the plot of the equivalent conductivity against the inverse of viscosity for MDESs over a temperature range from 25 to 75 °C is shown in Fig. 5d. When the conductivity and viscosity comply with Walden's rule, the relationship could be expressed with eqn (9):

$$\Lambda_\eta^\alpha = C \quad (9)$$

where α is the slope of the line in the Walden plot which reflects the decoupling of ions, and C is a temperature-dependent constant. The dotted ideal line in the plot represents the data for 0.01 mol L^{-1} KCl aqueous solution at high dilution, which is in accordance to a system with equal ion movement and dissociation.[49] As shown in Fig. 5d, almost all MDESs in this work are close to the reference line, which implies they could be regarded as good ILs. For the LA-based MDESs, they behave with a poorer performance than others. The deviations between MDESs and ideal solution derive from the different electrostatic interactions between HBA and HBD.[49] The DESs of ChCl:EG and ChCl:LA have been reported as electrolytes for lithium-ion batteries owing to their high ionic conductivity.[50] Comparing their ionicity at 25 °C, the values for LA-based MDESs nearly approach those for ChCl:LA and other MDESs have similar ionicity to ChCl:EG. When comparing with ionic species, such as [EMIM][BF_4] and TBAC:EG,[51,52] the MDESs could still match them. From the above results, the prepared MDESs have good conductivity behavior.

pH and surface tension

pH is an important physical property and it is basic for DESs to apply in catalysis, metal treatment, gas absorption and biomass dissolution. Thus, the pH values of these MDESs were measured at room temperature and the results are recorded in Table 2. From the table, it can be seen that the pH values range from 0.4 to 3.9, demonstrating that all MDESs behave with an acidic characteristic. Among them, the LA-based MDESs exhibit a stronger acidity than the others which might be due to the existence of a carboxylic acid group. Moreover, when using the same HBD, the MDESs have similar pH values, which resembles the same system with different HBD content as well (see Table S2, ESI†). Thus, the HBD components might have an effect on the pH behavior of MDESs which is consistent with a previous report.[53]

Surface tension is an energy which actually requires an increase in the surface area of a liquid by a unit of area. It can be used to understand the intermolecular forces between molecules in the mixture.[54] It is also an essential physical factor for industrial processes and chemical applications. In this study, we measured the surface tension of all MDESs at 25 °C. As is shown in Table 2, the surface tension values are located between 35.0 and 89.9 mN m^{-1}. On account of the higher viscosity, MDES2 has no surface tension data to record. When using the same HBD components, the Ni-based MDESs always have higher surface tension values than Co-based MDESs. This might be owing to the stronger H-bonding interaction between HBD and $NiCl_2{\cdot}6H_2O$ than $CoCl_2{\cdot}6H_2O$. The results indicate that HBA is an influencing factor to surface tension. For the MDESs with the same HBA, the surface tension order follows amide-based MDESs > alcohol-based MDESs > carboxylic acid-based MDESs in this work. Moreover, the MDESs have similar surface tension values to common DESs at 25 °C, such as ChCl:EG (49.4 mN m^{-1}), ChCl:LA (44.4 mN m^{-1}), $FeCl_3{\cdot}6H_2O$:EG (57.0 mN m^{-1}) and so on.[55] And the majority of typical ILs have surface tension values between 40 and 50 mN m^{-1}.[56] Thus, the MDESs are comparable to common DESs and ILs.

Conclusions

In this work, eleven novel magnetic eutectic solvents based on $CoCl_2{\cdot}6H_2O$/$NiCl_2{\cdot}6H_2O$ and three different types of HBD components including alcohols, carboxylic acids and amides were prepared successfully. The intermolecular interactions of the prepared MDESs and their deep eutectic features were

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characterized by FTIR, ^{1}H NMR and DSC analysis. In addition, the solvatochromic parameters, pH and surface tension values were measured at room temperature. All MDESs exhibit an acidic characteristic and have stronger H-bonding interaction networks. Their physicochemical properties including density, viscosity and conductivity were investigated as a function of temperature. In view of the stronger H-bonding interaction between $NiCl_2{\cdot}6H_2O$-based MDESs, they have a higher density, viscosity, surface tension and polarity values than $CoCl_2{\cdot}6H_2O$-based MDESs when combining the same HBD. And the HBD components also influence the physicochemical properties of different MDESs. A Walden plot was further applied to evaluate the ionicity of these MDESs. Almost all MDESs approach the ideal line, with good conductivity comparable with other DESs and ILs. At the meantime, the prepared MDESs behave a good thermostability. This work could help us better comprehend the behaviors of Co/Ni-based DESs, thereby benefiting the improvement of their usage and developing potential applications.

Author contributions

Ruifen Shi: investigation, writing – original draft, writing – review & editing. Fengyi Zhou, Shuzi Liu: investigation. Yu Chen, Zhenghui Liu, Tiancheng Mu: conceptualization, writing – review & editing, supervision.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

The authors thank the National Natural Science Foundation of China (22073112 and 22103030) for financial support.

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